Fever of Unknown Origin (FUO) & Travel Medicine, Part 1: An Approach to a Case of Fever

Dr. Majed Alnabulsi, Internal Medicine | Source: `FUOTravel_medicine_2.0.pdf` (42 slides)

Legend: ⭐ = high-yield · *[added]* = not on slides. Images marked "Slide N" come from the lecture; "Added diagram" = drawn for these notes. The lecture has **no answer key** for its cases; the answers below are **[reasoning added; key not on slides]**.

🔗 **Bridge from earlier lectures:**

- **Tuberculosis (Kotb):** TB, especially **extrapulmonary**, is the classic infectious cause of FUO. A **diagnostic trial of anti-TB therapy** is a last-resort FUO step. IGRA/TST appear in the obligatory FUO work-up.
- **Infective Endocarditis (Reihan):** IE is on every FUO list. **Take blood cultures (n = 3) before any antibiotic**; empiric antibiotics are the commonest reason IE becomes "culture-negative".
- **Cardiogenic shock / Resp failure–ARDS:** severe falciparum malaria → **shock/DIC** and **pulmonary oedema → ARDS**.
- **Psychotropics / psychiatry: mefloquine** is avoided in psychiatric illness and seizures.
- **Pneumonia (Kotb):** fever with relative bradycardia also points to atypical pneumonia (*Legionella*). Typhoid is the travel-medicine version.

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1. Title / Objectives

Domain	Objective	Likely tested as
Knowledge	Clinical manifestations and objective findings of FUO and travel illnesses	Classic FUO definition; 4 FUO categories
Knowledge	⭐ Epidemiology, manifestations, complications, management	Malaria species, relapse, G6PD, artesunate; typhoid; traveller's diarrhoea
Skill	Interpret manifestations and investigations	Blood smear; blood vs stool culture timing
Skill	Apply an evidence-based management plan	Avoid empiric therapy in FUO; species-specific antimalarials
Value	Professional attitude	—

Topics (slide 3): FUO → Malaria → Typhoid & Paratyphoid → Traveller's diarrhoea → generic "Approach to fever" (slides 31–34, repeated verbatim in Part 2).

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2. Main Content (Lecture Order)

2.0 Opening Case (Slide 5 = Case Scenario 1, slide 37)

Presentation:

- **29-year-old woman**, returned from **rural India** 1 week ago (NGO house building)
- **chloroquine prophylaxis**: 1 dose a week before the trip, weekly after; last dose 2 days ago
- high fever, headache, myalgia
- **T 39.3 °C**, HR 102, RR 22, BP 112/78
- **jaundice**, soft non-tender abdomen
- **Hb 10 g/dL**
- smear: **fragmented RBCs + ring-shaped inclusions within RBCs**

Most likely organism? → ⭐ **B. *Plasmodium falciparum*** *[reasoning added; key not on slides]*:

- **intra-erythrocytic ring forms** = *Plasmodium*. This excludes **dengue, chikungunya and hepatitis A**, which are viruses without RBC inclusions. That answers the slide's own prompt: "which organism this **can't** be".
- **India = chloroquine-resistant *P. falciparum***, so chloroquine prophylaxis fails. Slide 23: chloroquine works only in Central America, Haiti and parts of the Middle East.
- **haemolysis** (anaemia, jaundice, fragmented RBCs) with high fever → falciparum, the species with high parasitaemia
- ***P. ovale*** is chloroquine-sensitive and usually milder (tertian, "usually less severe", slide 14)

2.1 FUO: Definitions (Slide 6) ⭐⭐

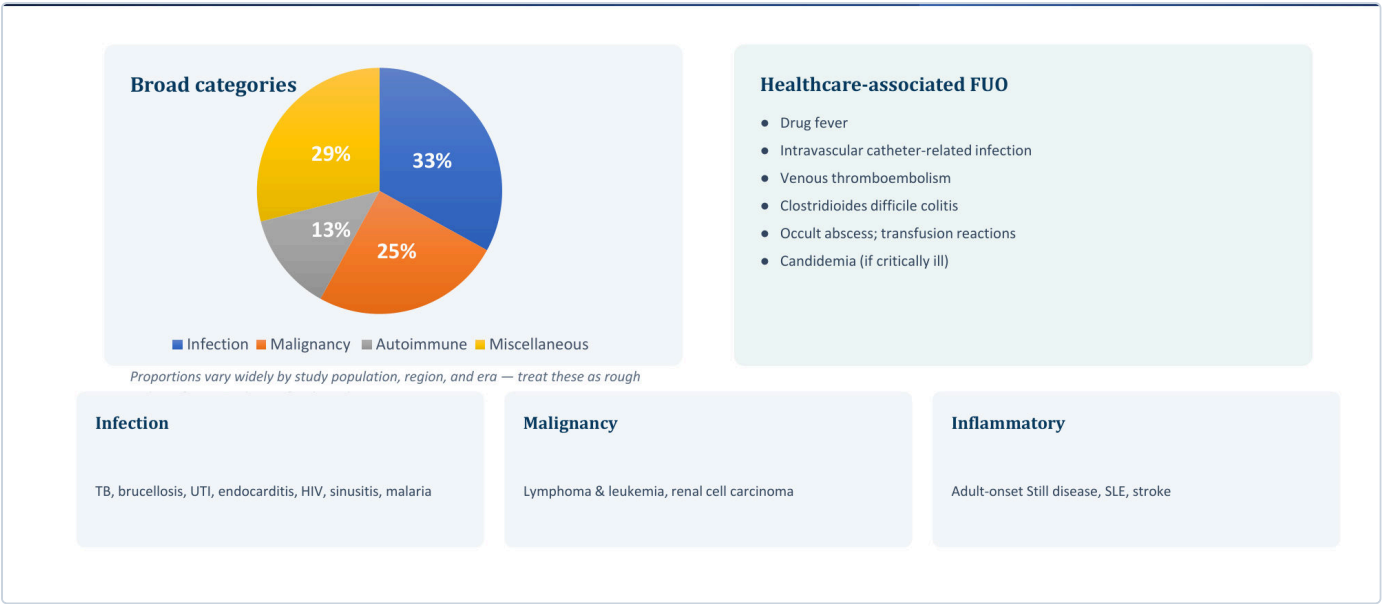
⭐ **Classic FUO** has three parts:

1. **temperature > 38.3 °C** on **multiple occasions**
2. lasting **> 3 weeks**
3. **no diagnosis despite 3 outpatient visits or 3 days in hospital**

Category	Definition (slide 6)
Classic	> 38.3 °C, > 3 weeks, undiagnosed after **3 outpatient visits or 3 inpatient days**
Nosocomial	Hospitalised; **no infection present or incubating at admission**; unexplained after **3 days** of evaluation
Neutropenic	**ANC < 500/mm³** (or expected to fall below); unexplained after **3 days, including 2 days of culture incubation**
HIV-associated	Confirmed HIV; fever **> 4 weeks (outpatient)** or **> 3 days (inpatient)**; unexplained after appropriate work-up

> ⚠️ **Slide inconsistency:** slide 6 writes "38.3 °C (100.1 °F)". 38.3 °C = **100.9 °F (≈ 101 °F)**, which is what the flowchart on slide 10 uses ("≥ 38.3 °C (≥ 101 °F)"). Learn **38.3 °C / 101 °F**.

2.2 FUO: Aetiology (Slides 7–8) ⭐



Slide 7: FUO aetiology categories (pie) and healthcare-associated causes

Pie chart:

- **Infection 33 %**
- **Malignancy 25 %**
- **Autoimmune 13 %**
- **Miscellaneous 29 %**

The slide warns these proportions **vary widely by population, region and era**, so treat them as rough orders of magnitude.

Category	Slide examples
Infection	**TB, brucellosis**, UTI, **endocarditis**, HIV, sinusitis, malaria
Malignancy	**Lymphoma & leukaemia**, **renal cell carcinoma**
Inflammatory	**Adult-onset Still disease**, SLE, "stroke"
Healthcare-associated FUO	**Drug fever**; intravascular **catheter infection**; **VTE**; ***C. difficile* colitis**; occult abscess; **transfusion reactions**; **candidaemia** (if critically ill)

⚠️ **Slide inconsistencies:**

- **"Stroke"** is listed under *inflammatory* causes. It is not inflammatory, though central fever after stroke does occur. It may be a misplaced item.
- The pie says "Autoimmune" but the box below says "Inflammatory". They mean the same group.

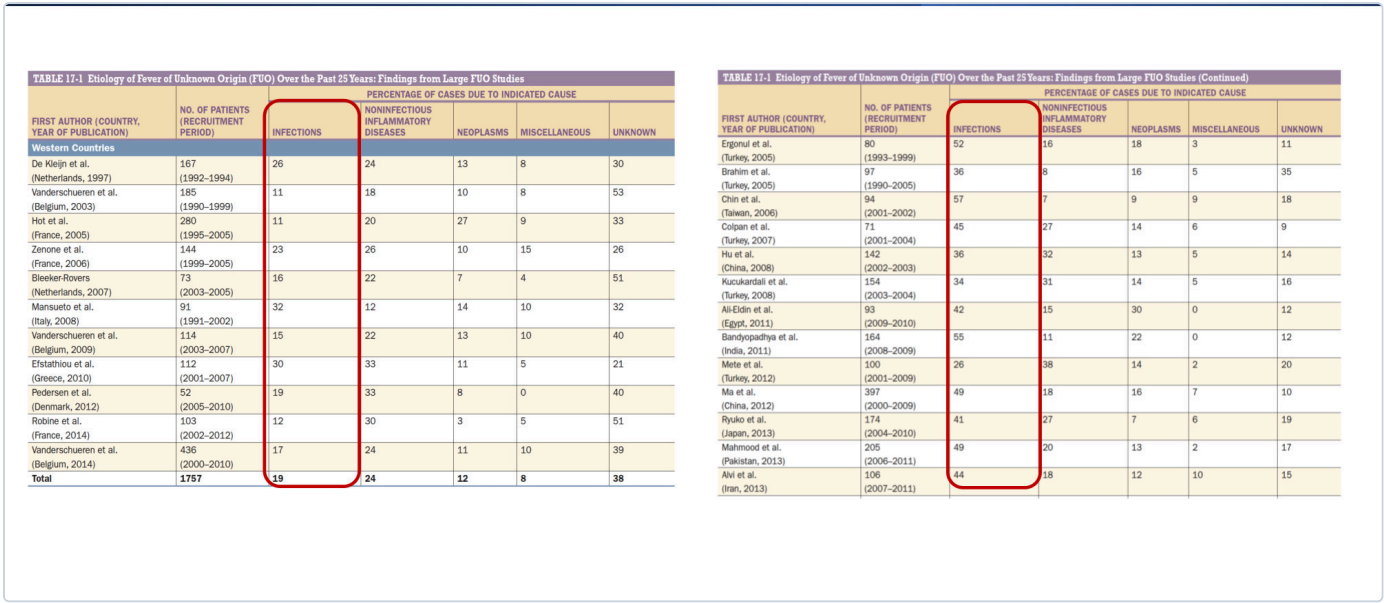


TABLE 17-1 Etiology of Fever of Unknown Origin (FUO) Over the Past 25 Years: Findings from Large FUO Studies

FIRST AUTHOR (COUNTRY, YEAR OF PUBLICATION)	NO. OF PATIENTS (RECRUITMENT PERIOD)	PERCENTAGE OF CASES DUE TO INDICATED CAUSE				
		INFECTIONS	NONINFECTIOUS INFLAMMATORY DISEASES	NEOPLASMS	MISCELLANEOUS	UNKNOWN
Western Countries						
De Kleijn et al. (Netherlands, 1997)	167 (1992–1994)	26	24	13	8	30
Vanderschueren et al. (Belgium, 2003)	185 (1990–1999)	11	18	10	8	53
Hot et al. (France, 2005)	280 (1995–2005)	11	20	27	9	33
Zenone et al. (France, 2006)	144 (1999–2005)	23	26	10	15	26
Bleeker-Rovers (Netherlands, 2007)	73 (2003–2005)	16	22	7	4	51
Mansueto et al. (Italy, 2008)	91 (1991–2002)	32	12	14	10	32
Vanderschueren et al. (Belgium, 2009)	114 (2003–2007)	15	22	13	10	40
Efstathiou et al. (Greece, 2010)	112 (2001–2007)	30	33	11	5	21
Pedersen et al. (Denmark, 2012)	52 (2005–2010)	19	33	8	0	40
Robine et al. (France, 2014)	103 (2002–2012)	12	30	3	5	51
Vanderschueren et al. (Belgium, 2014)	436 (2000–2010)	17	24	11	10	39
Total	**1757**	**19**	**24**	**12**	**8**	**38**

TABLE 17-1 Etiology of Fever of Unknown Origin (FUO) Over the Past 25 Years: Findings from Large FUO Studies (Continued)

FIRST AUTHOR (COUNTRY, YEAR OF PUBLICATION)	NO. OF PATIENTS (RECRUITMENT PERIOD)	PERCENTAGE OF CASES DUE TO INDICATED CAUSE				
		INFECTIONS	NONINFECTIOUS INFLAMMATORY DISEASES	NEOPLASMS	MISCELLANEOUS	UNKNOWN
Ergonul et al. (Turkey, 2005)	80 (1993–1999)	52	16	18	3	11
Brahim et al. (Turkey, 2005)	97 (1990–2005)	36	8	16	5	35
Chin et al. (Taiwan, 2006)	94 (2001–2002)	57	7	9	9	18
Colpan et al. (Turkey, 2007)	71 (2001–2004)	45	27	14	6	9
Hu et al. (China, 2008)	142 (2002–2003)	36	32	13	5	14
Kucukardali et al. (Turkey, 2008)	154 (2003–2004)	34	31	14	5	16
Ali-Eldin et al. (Egypt, 2011)	93 (2009–2010)	42	15	30	0	12
Bandyopadhya et al. (India, 2011)	164 (2008–2009)	55	11	22	0	12
Mete et al. (Turkey, 2012)	100 (2001–2009)	26	38	14	2	20
Ma et al. (China, 2012)	397 (2000–2009)	49	18	16	7	10
Ryuko et al. (Japan, 2013)	174 (2004–2010)	41	27	7	6	19
Mahmood et al. (Pakistan, 2013)	205 (2006–2011)	49	20	13	2	17
Alvi et al. (Iran, 2013)	106 (2007–2011)	44	18	12	10	15

Slide 8: Aetiology across large FUO series (Harrison's Table); infection column highlighted

- **Slide 8** shows Harrison's table of FUO series (1990s–2010s, Western countries on the left, other countries on the right).
- The **infection column is circled**: infection is a **larger share in non-Western series**.
- **"Undiagnosed"** is a large group in recent Western series *[reading of table; exact numbers too small to transcribe reliably]*.

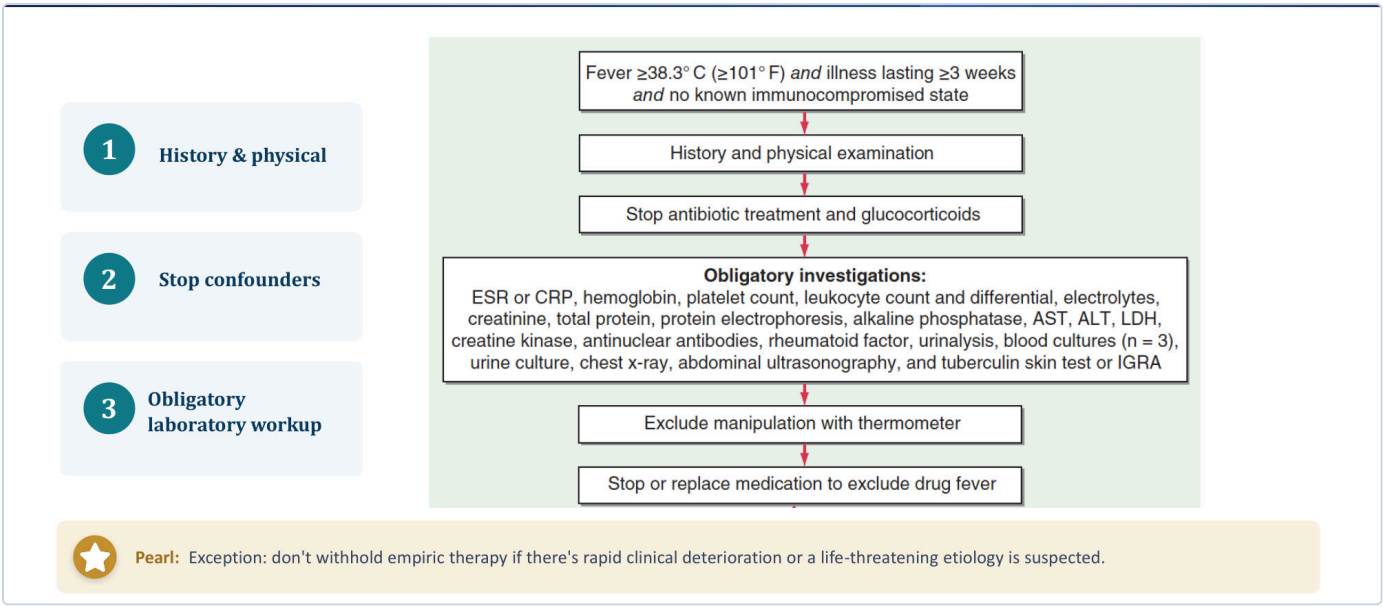
2.3 FUO: Clinical Features (Slide 9)

These features accompany fever without pointing to one cause:

Feature	Definition
Chills	Subjective cold, often with shivering
Rigors	**Severe chills** with pronounced shivering and **teeth chattering**
Night sweats	**Drenching**, soaking clothes or bedding
Change in mental status	Especially at the **extremes of age**

⭐ **Pearl:** look for **systemic clues to the cause**: skin changes, constitutional symptoms, **anaemia**, **unintentional weight loss**.

2.4 FUO: Patient Approach (Slides 10–11) ⭐⭐

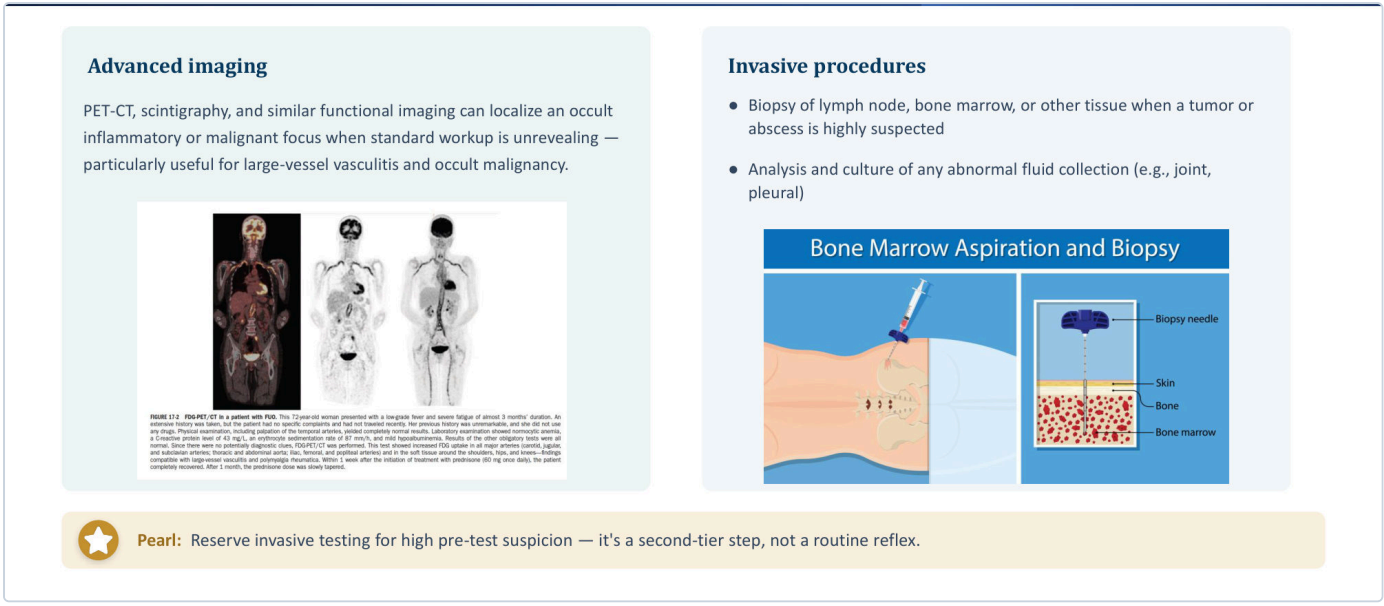


Slide 10: FUO approach flowchart

Flowchart, in order:

1. **Fever ≥ 38.3 °C (≥ 101 °F), illness ≥ 3 weeks, no known immunocompromise**
2. **History & physical examination**
3. ⭐ **Stop antibiotics and glucocorticoids** (the "confounders")
4. ⭐ **Obligatory investigations:**
   - **ESR or CRP**
   - Hb, platelets, **WBC with differential**
   - electrolytes, creatinine, total protein, **protein electrophoresis**
   - ALP, AST, ALT, **LDH**, **creatine kinase**
   - **ANA, rheumatoid factor**
   - urinalysis, **blood cultures (n = 3)**, urine culture
   - **chest X-ray**, **abdominal ultrasound**
   - **TST or IGRA**
5. **Exclude manipulation with the thermometer** (factitious fever)
6. **Stop or replace medication to exclude drug fever**

⭐ **Exception:** do **not** withhold empiric therapy if there is **rapid clinical deterioration** or a **life-threatening cause** is suspected.



Slide 11: Second-tier tests: PET-CT and bone-marrow biopsy

Second tier, if the work-up is unrevealing:

- **Advanced imaging:**
  - **PET-CT**, scintigraphy and similar functional imaging localise an occult inflammatory or malignant focus
  - especially useful for **large-vessel vasculitis** and **occult malignancy**
- **Invasive procedures:**
  - **biopsy** of lymph node, **bone marrow** or other tissue when tumour or abscess is strongly suspected
  - **analysis and culture of any abnormal fluid** (joint, pleural)
- ⭐ **Pearl:** invasive testing is reserved for **high pre-test suspicion**. It is a second-tier step, not a reflex.

2.5 FUO: Management (Slide 12) ⭐

Principle	Detail
Avoid empiric therapy	**Antibiotics and glucocorticoids mask the disease** and delay diagnosis
Know the exception	Treat empirically only with **rapid deterioration** or a **suspected life-threatening cause**
Treat the cause	Once it is found
Still undiagnosed?	**Specialist consultation**

⭐ **Pearl:** a specialist may try a **diagnostic trial** of:

- **anti-TB therapy**
- **colchicine** *[added: FMF]*
- **steroids**
- **anakinra (IL-1 receptor antagonist)** *[added: autoinflammatory disease, Still disease]*

[added] **Prognosis:** FUO that stays undiagnosed after a thorough work-up generally has a **good prognosis**.

> 🔁 **Recap: FUO.**
>
> - **Definition:** > 38.3 °C, > 3 weeks, undiagnosed after 3 visits or 3 inpatient days. Three other categories: nosocomial, neutropenic (ANC < 500), HIV.
> - **Causes:** infection > misc > malignancy > autoimmune.
> - **Approach:** stop antibiotics and steroids → obligatory panel (incl. **3 blood cultures**, **TST/IGRA**) → exclude **factitious** and **drug fever** → PET-CT → biopsy.
> - **Treatment:** no empiric treatment unless deteriorating.
>
> 🔗 **Link:** malaria is on the FUO infection list and is the first travel-medicine illness. Any fever in a traveller from an endemic area is **malaria until proven otherwise** *[added]*.
>
> ▶️ **Up next: Malaria.**

2.6 Malaria: Overview & Aetiology (Slides 13–14) ⭐

- **Protozoal** infection transmitted by the bite of an infected **female *Anopheles*** mosquito.
- **Where:** tropics and subtropics, especially **sub-Saharan Africa**, **South and Southeast Asia**, parts of the **Middle East** and **South America**.
- **Burden: ~212 million cases/year**, **~429,000 deaths/year**.
- **Vaccine:** "No vaccine widely deployed historically"; antimalarial **resistance is increasing**.
  - *[added]* WHO now recommends RTS,S and R21 for children in endemic Africa. There is still no vaccine for travellers.

- caused by one of five plasmodium parasites:



Classic disease and fever patterns of different plasmodium species [9][10].

Different species of plasmodium [11][12]	Disease	Fever spikes
P. vivax P. ovale	• Tertian malaria (usually less severe)	• Every 48 hours
P. malariae	• Quartan malaria (usually less severe)	• Every 72 hours
P. falciparum	• Falciparum malaria (most severe form; also known as malignant tertian malaria)	• Irregular
P. knowlesi	• Quotidian malaria	• Every 24 hours

Pearl: P. vivax and P. ovale carry a dormant liver stage (hypnozoites) that the other species lack — the key to understanding relapse, coming up next.

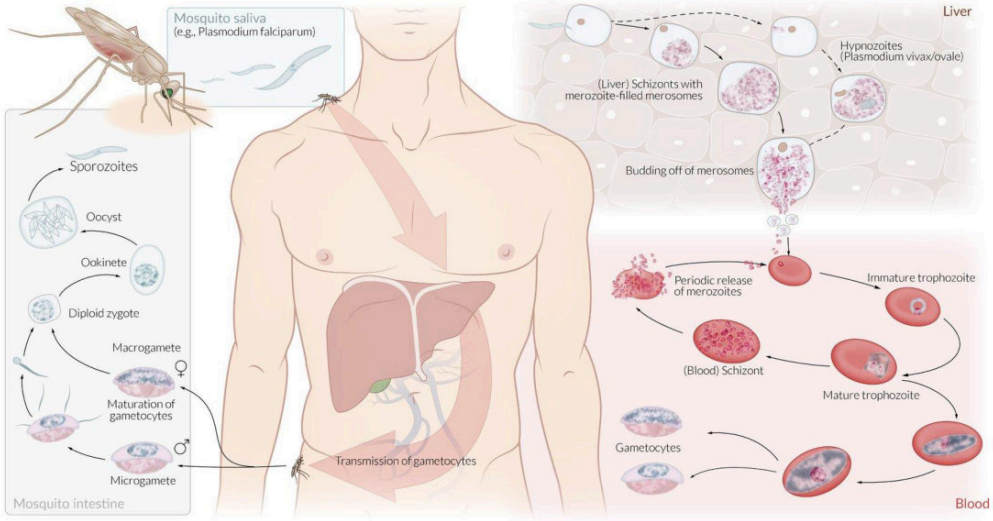
Slide 14: Five Plasmodium species: disease and fever pattern

Species	Disease	Fever spikes
P. vivax, P. ovale	**Tertian** malaria (usually less severe)	**Every 48 h**
P. malariae	**Quartan** malaria (usually less severe)	**Every 72 h**
P. falciparum	**Falciparum malaria**: **most severe**; "malignant tertian"	**Irregular**
P. knowlesi	**Quotidian** malaria	**Every 24 h**

⭐ **Pearl: *P. vivax* and *P. ovale* carry a dormant liver stage (hypnozoites)** that the other species lack. This is the basis of **relapse**.

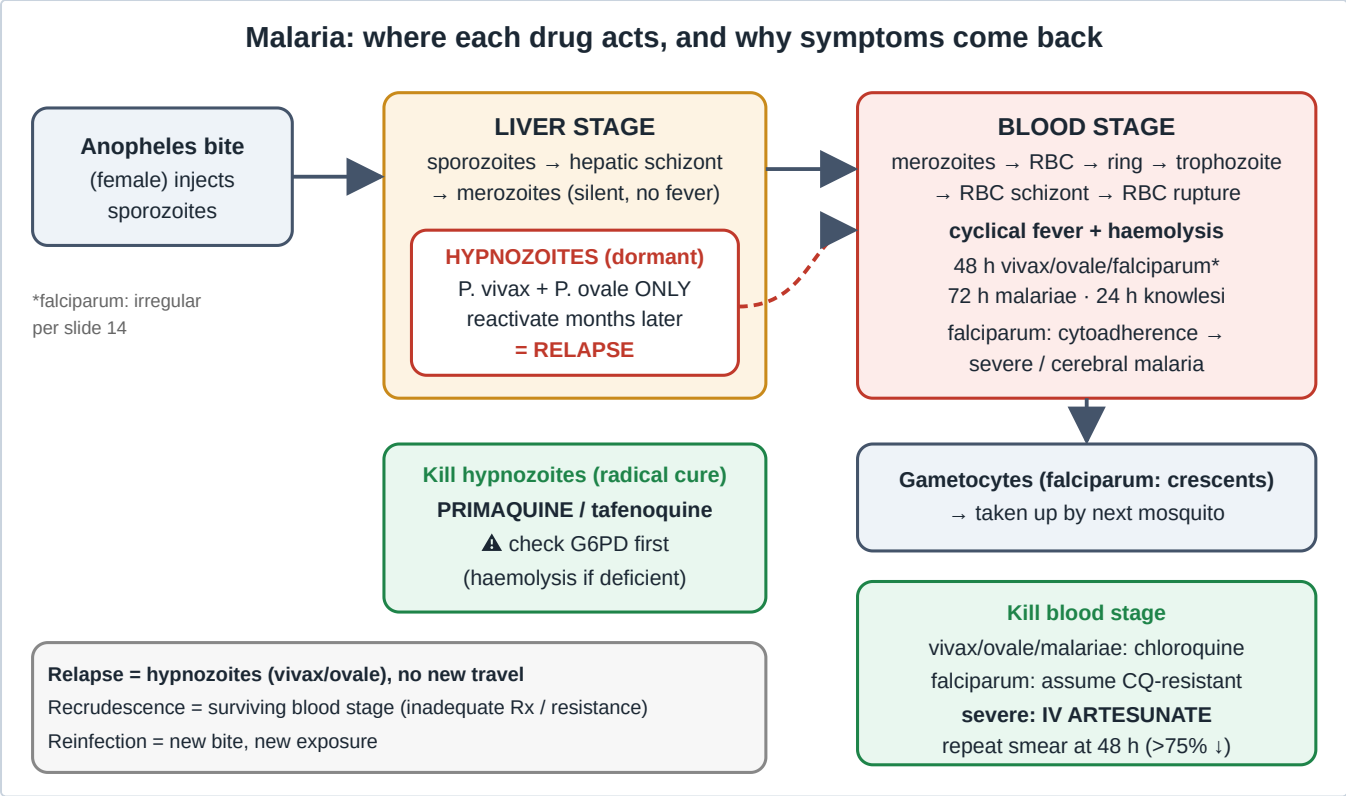
Mnemonic for fever periodicity: "Quartan = 4th day = 72 h; Tertian = 3rd day = 48 h; Quotidian = daily". Counting is inclusive: day 1 fever → day 3 fever = 48 h.

2.7 Malaria: Life Cycle (Slides 15–16) ⭐



Slide 15: Plasmodium life cycle (mosquito, liver, blood)

Step	Event
1. **Inoculation**	Anopheles injects **sporozoites** into the blood
2. **Liver stage**	Sporozoites invade **hepatocytes**, multiply asexually → **schizonts full of merozoites** (no symptoms)
3. **Blood stage**	Merozoites invade **RBCs** → **trophozoites** → **RBC schizonts**
4. **Haemolysis & symptoms**	**RBC rupture** releases more merozoites. This **cyclical rupture causes fever and haemolysis**
5. **Gametocytes**	Some merozoites become sexual **gametocytes**
6. **Back to the mosquito**	A new mosquito ingests gametocytes → they mature → **salivary glands** → next host



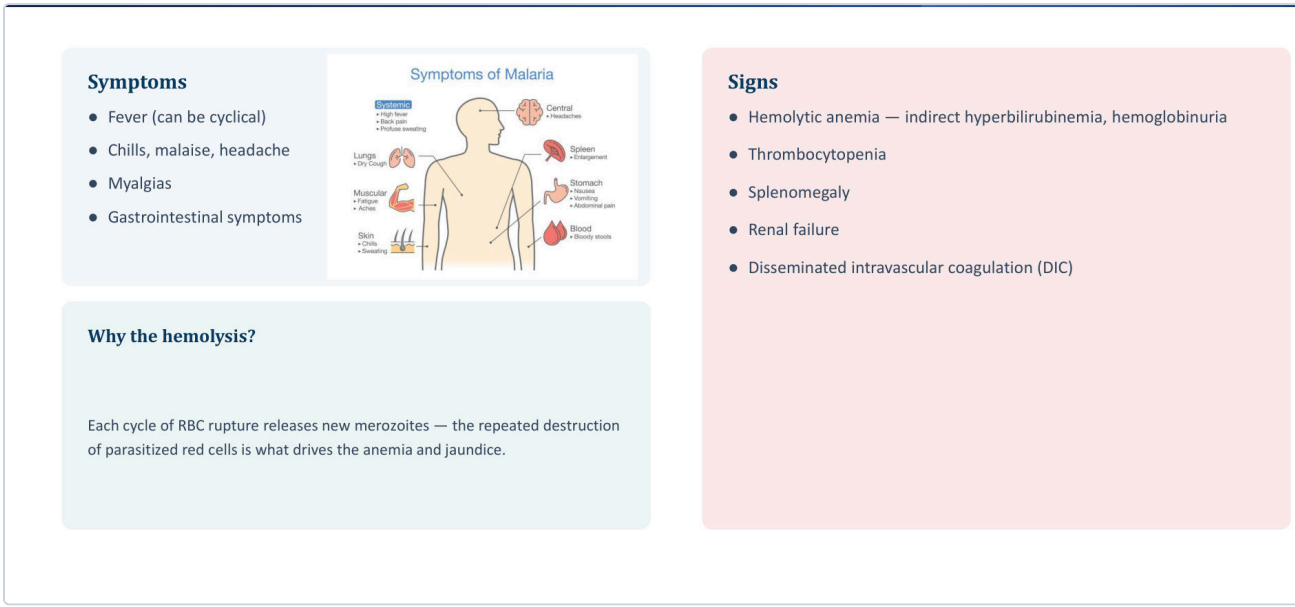
Added diagram: Malaria stages, drug targets, relapse vs recrudescence

2.8 Relapse vs Recrudescence vs Reinfection (Slide 17) ⭐⭐

	Relapse	Recrudescence	Reinfection
Mechanism	**Dormant liver hypnozoites reactivate** and re-seed the blood	**Blood-stage parasites persisted** at sub-microscopic levels and rebound	**New infection** from a new bite
Species	⭐ ***P. vivax* and *P. ovale* only**	Any; classically ***P. falciparum*** or ***P. malariae***	Any, with new exposure
Meaning / action	Needs **primaquine (or tafenoquine)** for radical cure; **check G6PD first**	**Inadequate treatment or drug resistance**	Distinguished by the **travel/exposure timeline**, not by testing alone

⭐ **Pearl:** a patient treated successfully for malaria months ago, with the same illness now and **no new travel** → **relapse** (hypnozoite reactivation), **not treatment failure**. This is Case 3.

2.9 Malaria: Clinical Features (Slide 18)



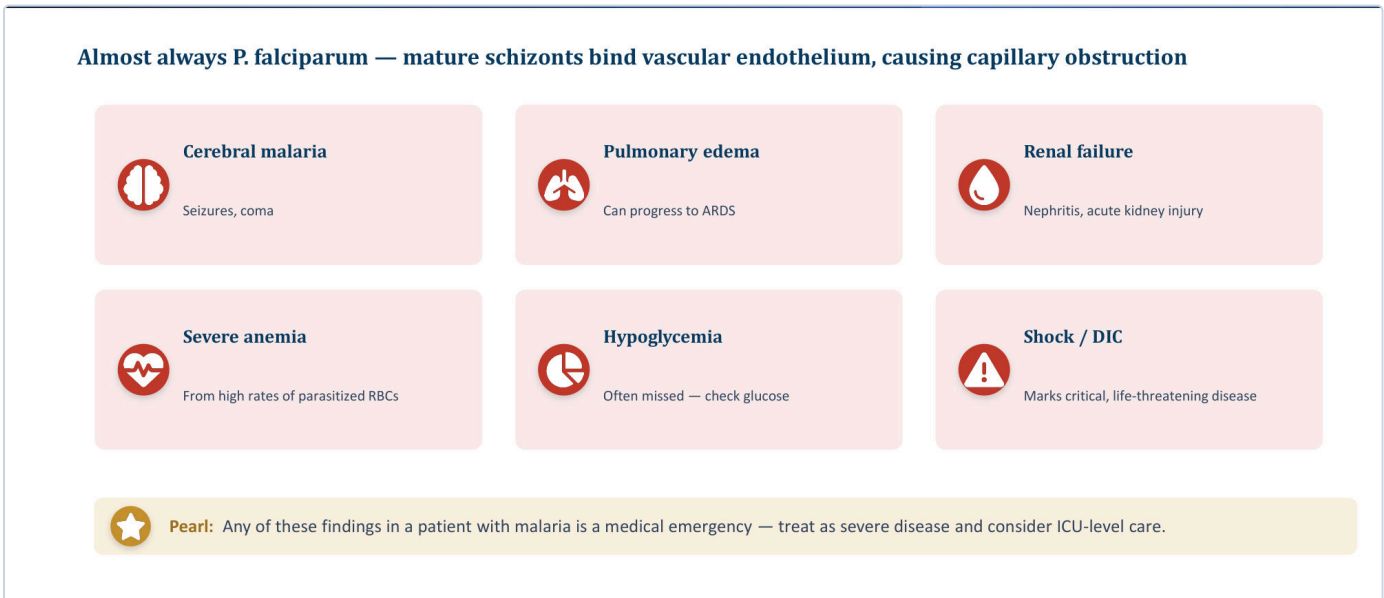
Slide 18: Malaria symptoms and signs

Symptoms	Signs
Fever (can be cyclical)	**Haemolytic anaemia**: **indirect** hyperbilirubinaemia, **haemoglobinuria**
Chills, malaise, headache	**Thrombocytopenia**
Myalgia	**Splenomegaly**
GI symptoms	**Renal failure**
	DIC

Why the haemolysis? Each cycle of RBC rupture releases new merozoites. **Repeated destruction of parasitised RBCs** drives the **anaemia and jaundice**.

2.10 Severe (Complicated) Malaria (Slide 19) ⭐⭐

⭐ **Almost always *P. falciparum*:** mature schizonts **bind vascular endothelium** (cytoadherence) → **capillary obstruction**.



Slide 19: Severe malaria complications

Complication	Detail
Cerebral malaria	**Seizures, coma**
Pulmonary oedema	Can progress to **ARDS**
Renal failure	Nephritis, AKI *[added: "blackwater fever" = massive haemolysis → haemoglobinuria]*
Severe anaemia	High parasitised-RBC rate
⭐ **Hypoglycaemia**	**Often missed: check glucose** *[added: worsened by quinine-induced insulin release]*
Shock / DIC	Critical, life-threatening

⭐ **Pearl: any one** of these = **medical emergency** → treat as severe disease and consider **ICU**.

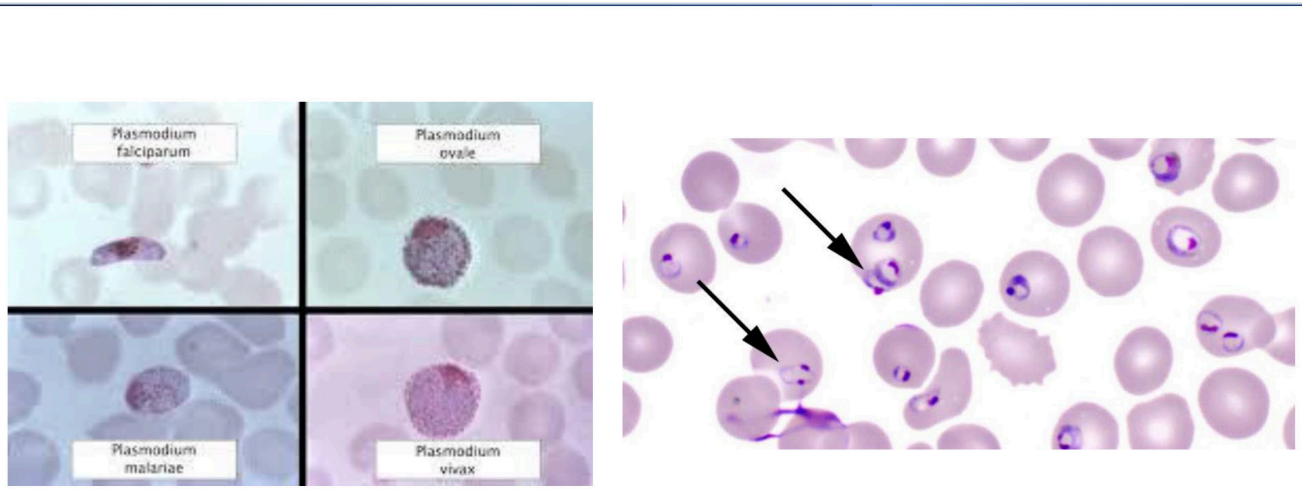
2.11 Malaria: Diagnostics (Slides 20–21) ⭐⭐

Purpose	Tests
Organ dysfunction	CBC, comprehensive metabolic panel, LFTs, **coagulation panel**
Parasitological	⭐ **Thick & thin blood smear (Giemsa/Wright)**, the **gold standard**. *[added: thick = detect; thin = species and % parasitaemia]*
	Rapid antigen test: faster, **less sensitive**

Smear clues:

- **Schüffner granules** (fine **brick-red dots**) in RBCs → ***P. vivax* and *P. ovale***
- **crescent-shaped gametocytes** → ***P. falciparum***
- *[added]* falciparum also shows multiple rings per cell, **appliqué/headphone** forms, and RBCs of normal size; vivax/ovale enlarge the RBC; malariae shows **band forms**

⭐ **Don't stop at one:** a **single negative smear does not rule out malaria**. Repeat **every 12–24 h for 3 sets** before excluding it.



Slide 21: Smears of the four classic species, plus falciparum ring forms (arrows)

2.12 Malaria: Management (Slide 22) ⭐⭐

Species	Treatment
P. vivax, P. ovale, P. malariae	**Chloroquine**
+ *vivax / ovale*	**Add primaquine** to eradicate **hypnozoites** and prevent relapse
	⚠️ **Check G6PD first**: primaquine → **severe haemolysis** in G6PD deficiency
P. falciparum	**Assume chloroquine resistance**
Uncomplicated (slide's options)	**Quinine + doxycycline**; **quinine + sulfadoxine/pyrimethamine**; **mefloquine**; **atovaquone/proguanil**
⭐ **Severe disease**	**IV artesunate** (artemisinin) is the **drug of choice**

⭐ **Monitoring:**

- repeat smears at **48 h**; expect **> 75 % fall in parasitaemia**
- consider **exchange transfusion** if parasitaemia **> 10–15 %**

> ⚠️ **Outdated on slide:** for **uncomplicated falciparum**, current WHO/CDC first line is an **artemisinin-based combination therapy (ACT)**, e.g. **artemether–lumefantrine** *[added]*. The slide's quinine/mefloquine options are alternatives. For the exam, know the slide list; in practice, ACT. Also *[added]*: chloroquine-resistant *P. vivax* exists (Indonesia/Oceania) → ACT.

2.13 Malaria: Prevention (Slides 23–24) ⭐

Bite avoidance first:

- **bed nets**, window screens, insecticides
- **repellent with 30–35 % DEET**

Destination	Chemoprophylaxis
Central America, Haiti, parts of the Middle East	**Chloroquine** (still effective there)
Most other endemic areas	**Mefloquine** or **atovaquone/proguanil**
Southeast Asia	**Doxycycline** or **atovaquone/proguanil** (resistance to others common)

Drug	Side effect / contraindication (slide)
Atovaquone/proguanil	Avoid in **renal disease** and **pregnancy**
Mefloquine	Avoid with **seizures** or **psychiatric conditions**
Chloroquine	**Well tolerated, safe in pregnancy**
Primaquine	**Contraindicated in G6PD deficiency** (haemolysis)

[added] **Timing:**

- **doxycycline**: start 1–2 days before, continue **4 weeks** after return; photosensitivity
- **mefloquine**: start ≥ 2 weeks before, continue 4 weeks after
- **atovaquone/proguanil**: start 1–2 days before, continue **7 days** after
- **chloroquine**: start 1–2 weeks before, continue 4 weeks after

- **mefloquine** is the option in **pregnancy** where there is chloroquine resistance

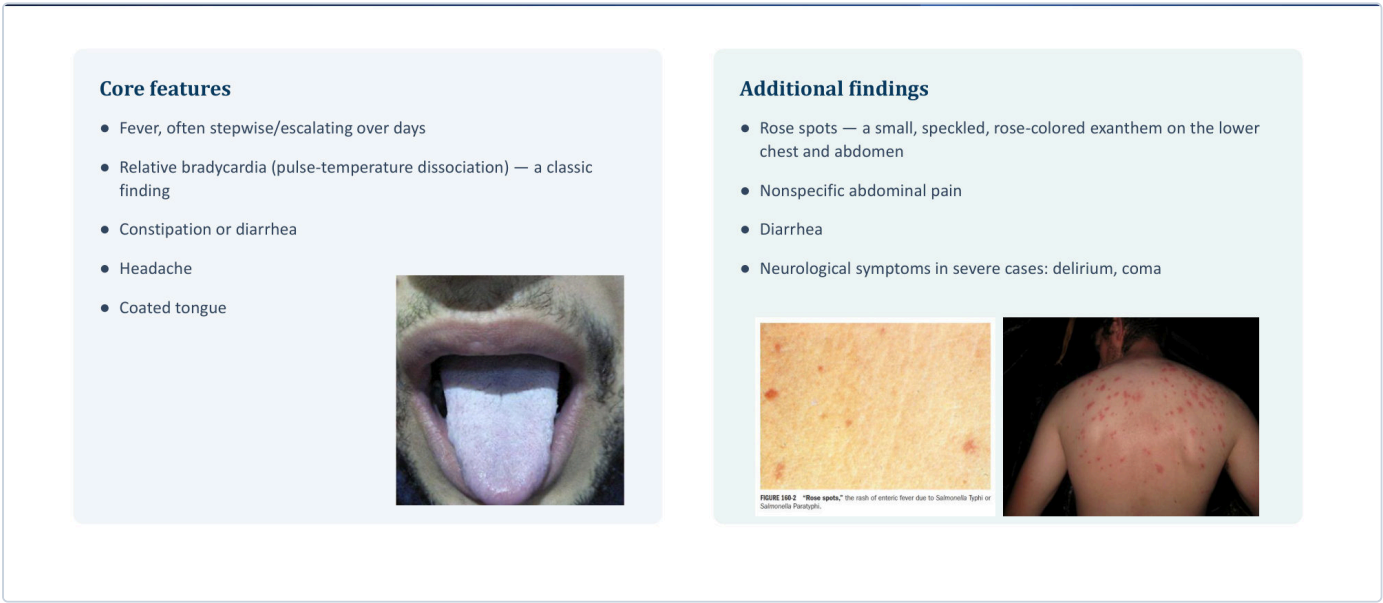
- **Slide 24** is a screenshot of the **CDC Travelers' Health** page ("Where are you going?").

- Message: check **destination-specific** advice before travel.

> 🔁 **Recap: Malaria.**
> - **Species & fever:** 5 species; vivax/ovale tertian 48 h, malariae quartan 72 h, falciparum irregular and severe, knowlesi 24 h.
> - **Relapse:** only vivax/ovale (hypnozoites) → **primaquine after G6PD**.
> - **Severe disease:** = falciparum (cerebral, ARDS, AKI, anaemia, **hypoglycaemia**, shock/DIC) → **IV artesunate**.
> - **Diagnosis:** thick/thin smear ×3.
> - **Prophylaxis:** chloroquine only in Central America/Haiti/parts of the Middle East.
>
> 🔗 **Link:** the next travel fever is also systemic and caught via **food/water** rather than a vector. It is the one with **relative bradycardia**.
>
> ▶️ **Up next: Typhoid.**

2.14 Typhoid & Paratyphoid: Overview & Clinical (Slides 25–26) ⭐⭐

- **Organism: *Salmonella enterica*** serotype **Typhi** and **Paratyphi (A, B, C)**.
- **Where:** poor sanitation: East and Southeast Asia, Africa, Central and South America.
- **Incubation: 5–30 days**.
- **Transmission: faecal–oral** (contaminated food and water).



Slide 26: Coated tongue and rose spots

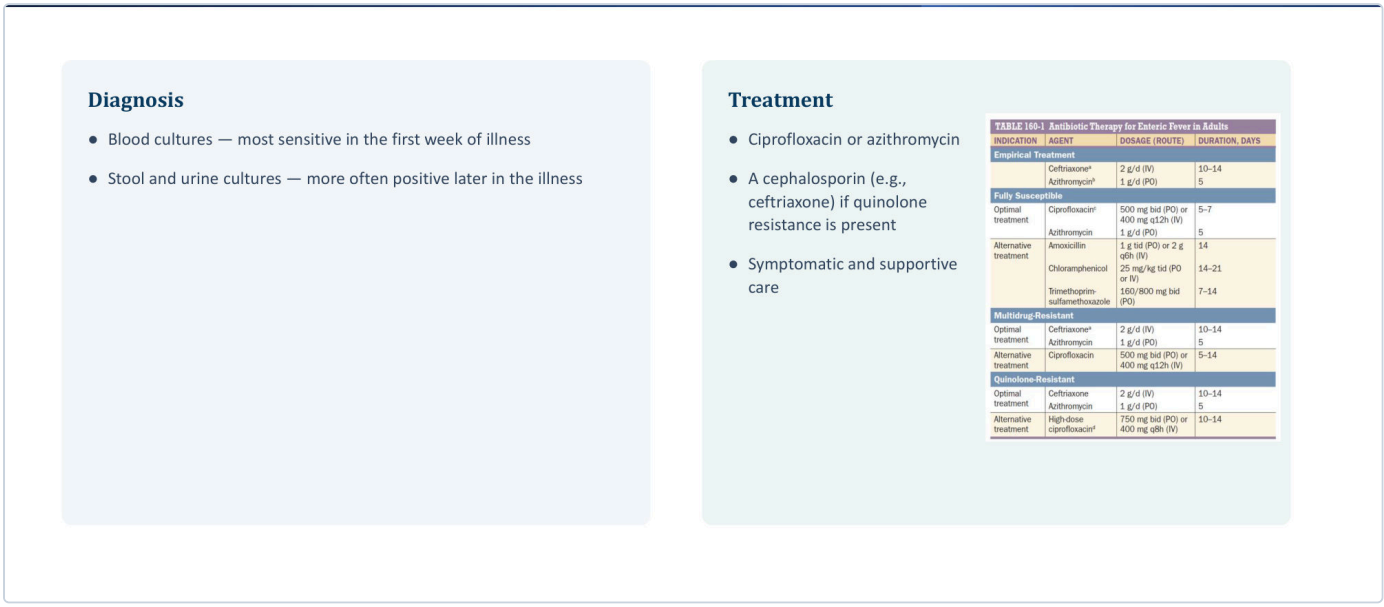
Core features	Additional findings
Fever, often **stepwise/escalating** over days	⭐ **Rose spots**: small, speckled, rose-coloured rash on the **lower chest and abdomen**
⭐ **Relative bradycardia** (pulse–temperature dissociation)	Non-specific abdominal pain
Constipation or diarrhoea	Diarrhoea
Headache	**Neurological** in severe cases: **delirium, coma**
Coated tongue	*[added: hepatosplenomegaly, leukopenia]*

[added] **Classic sequence:**

- week 1: **stepwise fever + constipation**

- week 2: **rose spots**, splenomegaly, "pea-soup" diarrhoea
- week 3: complications

2.15 Typhoid: Diagnostics & Management (Slide 27) ⭐



Slide 27: Typhoid antibiotic table (Harrison's Table 160-1, partly cropped on the slide)

Diagnosis:

- ⭐ **Blood cultures**: **most sensitive in week 1**
- **stool and urine cultures**: positive **later** in the illness
- *[added]* **bone-marrow culture** is the most sensitive overall, even after antibiotics
- *[added]* the **Widal** test is unreliable

Treatment:

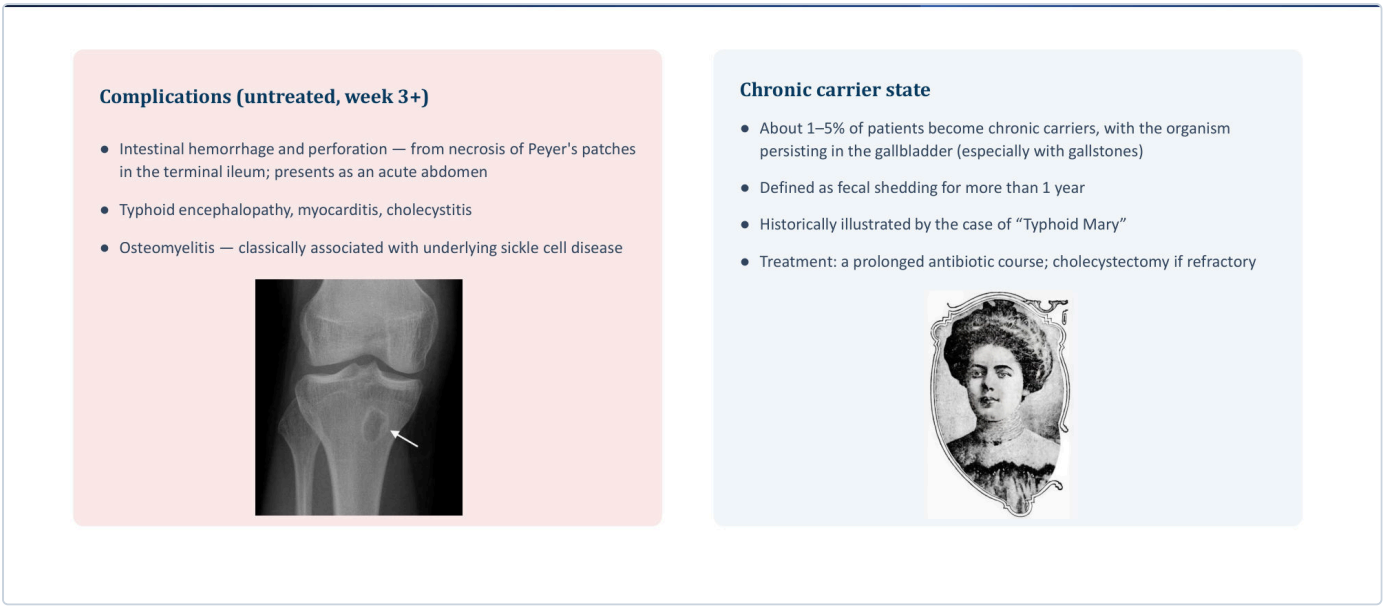
- **ciprofloxacin or azithromycin**
- **cephalosporin (ceftriaxone)** if **quinolone resistance**
- symptomatic and supportive care

Harrison's table on the slide:

Situation	Optimal	Alternative
Empirical	Ceftriaxone; azithromycin	—
Fully susceptible	Ciprofloxacin; azithromycin	Amoxicillin, chloramphenicol, TMP-SMX
Multidrug-resistant	Ceftriaxone; azithromycin	Ciprofloxacin
Quinolone-resistant	Ceftriaxone; azithromycin	High-dose ciprofloxacin

Doses are cropped off the slide.

2.16 Typhoid: Complications & Carrier State (Slide 28) ⭐



Slide 28: Osteomyelitis X-ray; "Typhoid Mary"

Complications (untreated, week 3+):

- ⭐ **Intestinal haemorrhage and perforation**: **necrosis of Peyer's patches in the terminal ileum** → **acute abdomen**
- typhoid **encephalopathy**, **myocarditis**, **cholecystitis**
- **osteomyelitis**: classically with **sickle cell disease**

Chronic carrier state:

- **1–5 %** of patients
- organism persists in the **gallbladder** (especially with **gallstones**)
- definition: **faecal shedding > 1 year**
- "**Typhoid Mary**"
- treatment: **prolonged antibiotics**; **cholecystectomy** if refractory

🔁 **Recap: Typhoid.**

- *S.* Typhi/Paratyphi, faecal–oral, incubation 5–30 d.
- **Clinical:** stepwise fever, **relative bradycardia**, **rose spots**, coated tongue, constipation → diarrhoea.
- **Diagnosis: blood culture in week 1**; stool/urine later.
- **Treatment:** cipro/azithro; ceftriaxone if quinolone-resistant.
- **Complications: Peyer's patch perforation** in week 3; carrier = gallbladder.

🔗 **Link:** both typhoid and traveller's diarrhoea are faecal–oral. Typhoid is **systemic/invasive**; traveller's diarrhoea is usually **toxin-mediated, watery, self-limiting**.

▶️ **Up next: Traveller's diarrhoea.**

2.17 Traveller's Diarrhoea (Slides 29–30) ⭐

Definition:

- sudden **≥ 3 loose/watery stools per day**
- **plus ≥ 1** GI or systemic symptom: fever, cramps, nausea, abdominal pain, or blood in stool

Onset:

- usually **within 1 week of arrival**
- watery diarrhoea lasting **2–4 days**

Aetiology	Note
⭐ **Enterotoxigenic *E. coli* (ETEC)**	**> 50 % of cases**
Campylobacter, *Shigella*, *Salmonella*	Invasive bacteria
Parasites: ***Giardia***, ***Entamoeba***, ***Cryptosporidium***	*[added: think parasites when diarrhoea lasts > 2 weeks]*

⭐ **Pearl: dysentery** (bloody diarrhoea + fever) → ***Shigella* or *Entamoeba***, not the typical toxin-mediated picture.

Risk & diagnostics	Management
Highest risk: **South & Southeast Asia, sub-Saharan Africa, Middle East, Latin America**	⭐ **Fluid replacement**: the mainstay
Depends on trip type/duration and host factors	Antibiotics have a **limited role**: **fluoroquinolone, azithromycin, or rifaximin** for more severe cases
Identifying the organism is usually unnecessary; work-up only for **systemic illness or prolonged disease**	Symptomatic: **loperamide** or **diphenoxylate** (antimotility)

[added]:

- avoid antimotility agents in **dysentery** (bloody, febrile)
- **azithromycin** is preferred for dysentery and in **Southeast Asia** (quinolone-resistant *Campylobacter*)
- **rifaximin** is for **non-invasive** diarrhoea only

2.18 Approach to Fever: Generic Framework (Slides 31–34)

These four slides are **repeated verbatim in Part 2**.

Key history	Key physical	General work-up	Specific work-up
Severity, **duration, pattern**	**Vital signs**	CBC, BMP, LFTs	**ANA, RF**
Localising symptoms (rash, wheeze, cough, ear discharge)	HEENT	UA, urine culture	**LP**
Night sweats, weight loss	Neck	CXR, sputum culture	**Serology**
⭐ **Travel history**	Heart (murmur → IE)	Stool analysis	**Abdominal imaging**
Sick contacts; animal contact	Lung	Viral antibodies	**TB** testing
Immunisations; **recent surgery**; **medications**	Abdomen, **skin**		

Slide 33 differential:

Infections	Malignancy	Autoimmune	Miscellaneous
TB (esp. extrapulmonary); abdominal / pelvic / dental **abscess**; **endocarditis**; **osteomyelitis**; sinusitis; **CMV, EBV, HIV**; Lyme; **prostatitis**	**Chronic leukaemia, lymphoma**, metastases, **RCC**, colon ca, **hepatoma**, MDS, pancreatic ca, sarcoma, (sarcoidosis)	**Adult Still's**, **PMR**, **temporal arteritis**, RA, rheumatic fever, **IBD**, reactive arthritis (Reiter's), **SLE**, vasculitides	**Drug fever**, cirrhosis complications, **factitious fever**, hepatitis (alcoholic, granulomatous, lupoid), **DVT**

> ⚠️ **Slide 33 errors:**
> - **"Sinusitis" listed twice**
> - **"Rheumatoid fever"** should read **rheumatic fever**
> - **sarcoidosis** sits under *malignancies* (it is granulomatous, not malignant)
> - slide 34: "**Urine culutres**" (typo)

2.19 To Summarise (Slide 35)

	Cause	Hallmark	Treatment
Malaria	*Plasmodium* spp. via *Anopheles*	**Cyclical fever, haemolytic anaemia, splenomegaly**	Species-specific; **artesunate for severe**
Typhoid	*S.* Typhi / Paratyphi	**Stepwise fever, relative bradycardia, rose spots**	**Cipro or azithro**
Traveller's diarrhoea	**ETEC** (mostly)	Watery diarrhoea **within 1 week of arrival**	**Fluids** ± antibiotic ± loperamide

Take-home points:

- only **vivax/ovale relapse**; check **G6PD** before primaquine
- a **single negative smear** doesn't exclude malaria
- **rose spots + relative bradycardia + endemic travel = typhoid**

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3. High-Yield Comparison Tables

Table A: The Five *Plasmodium* Species

[slides 14, 17, 20, 22 + added]

	falciparum	***vivax***	***ovale***	***malariae***	***knowlesi***
Fever cycle	**Irregular**	48 h	48 h	**72 h**	**24 h**
Severity	⭐ **Most severe**	Usually mild	Mild	Mild	Can be severe *[added]*
Hypnozoites / relapse	❌	✅	✅	❌	❌
Smear clue	**Crescent gametocytes**, multiple rings	**Schüffner dots**, enlarged RBC	**Schüffner dots**, oval RBC *[added]*	Band forms *[added]*	Resembles malariae *[added]*
Treatment	**Assume CQ-resistant**; **IV artesunate if severe**	**CQ + primaquine**	**CQ + primaquine**	**CQ**	*[added]* ACT

	falciparum	*vivax*	*ovale*	*malariae*	*knowlesi*
Recrudescence classic	✅	—	—	✅ (can persist years)	—

Table B: Malaria Prophylaxis Drug Selection

Drug	Where	Avoid in
Chloroquine	Central America, Haiti, parts of Middle East	(well tolerated; **safe in pregnancy**)
Mefloquine	Most other areas	**Seizures, psychiatric illness**
Atovaquone/proguanil	Most areas incl. SE Asia	**Renal disease, pregnancy**
Doxycycline	**SE Asia**	*[added]* pregnancy, children < 8 y
Primaquine	(radical cure)	**G6PD deficiency**

Table C: Typhoid vs Traveller's Diarrhoea vs Malaria

	Malaria	Typhoid	Traveller's diarrhoea
Route	**Anopheles bite**	**Faecal–oral**	**Faecal–oral**
Incubation	*[added]* ≥ 7 days (falciparum usually < 1 month; vivax/ovale up to months–years)	**5–30 days**	**< 1 week of arrival**
Key sign	Cyclical fever, haemolysis, splenomegaly	**Relative bradycardia, rose spots**	Watery diarrhoea, 2–4 days
Diagnosis	**Thick/thin smear ×3**	**Blood culture (week 1)**	Usually none
Treatment	Species-specific	Cipro/azithro; ceftriaxone if resistant	**Fluids**

Table D: FUO Categories

See §2.1. Memorise **3 weeks / 3 visits / 3 days**; **ANC < 500 + 2 days culture**; **HIV: > 4 weeks outpatient, > 3 days inpatient**.

Table E: Culture Timing in Typhoid

Week	Best specimen
1	**Blood**
2–3	**Stool, urine**
Any (incl. after antibiotics)	*[added]* **Bone marrow**

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4. Rules, Exceptions, Patterns

Rule	Exception / nuance
FUO → **no empiric antibiotics/steroids**	**Rapid deterioration or life-threatening cause** → treat
Stop antibiotics and steroids before the FUO work-up	—
Malaria relapse = **vivax/ovale only**	Recurrence with *falciparum/malariae* = **recrudescence** or **reinfection**

Rule	Exception / nuance
Primaquine for radical cure	**Check G6PD first**
Chloroquine for non-falciparum	***P. falciparum*: assume resistance**
Severe malaria ≈ falciparum	*[added] knowlesi* (and occasionally *vivax*) can be severe
Single negative smear	**≠ exclusion**; 3 sets, 12–24 h apart
Typhoid: **high fever + tachycardia expected**	**Relative bradycardia** is the classic exception
Traveller's diarrhoea: antibiotics rarely needed	**Severe / dysentery / systemic** → azithro, FQ, rifaximin
Loperamide for symptom relief	*[added]* **Not in dysentery**
Chloroquine prophylaxis	Works only in **Central America, Haiti, parts of the Middle East**

Pattern recognition:

- **traveller + fever + haemolysis/jaundice + thrombocytopenia → malaria**
- **traveller + stepwise fever + relative bradycardia + rose spots + constipation → typhoid**
- **fever months after treated malaria, no new travel → vivax/ovale relapse**
- **sickle cell + osteomyelitis → *Salmonella***
- **gallstones + chronic shedding → typhoid carrier**

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5. Clinical Correlations

- **Any fever in a returning traveller → think malaria first.** Smear ×3 even if the patient took prophylaxis (Case 1: chloroquine prophylaxis in India failed).
- **Hypoglycaemia** in severe malaria: check glucose in any confused or comatose patient. *[added]* Quinine worsens it.
- **G6PD deficiency** is common in the Middle East. *[added]* Always test before primaquine or tafenoquine; G6PD deficiency is prevalent in Saudi Arabia.
- **Typhoid perforation** presents as an acute abdomen in week 3 → surgical review.
- **Sickle cell patients** → *Salmonella* osteomyelitis.
- **Factitious fever:**
  - "exclude thermometer manipulation" is an obligatory FUO step
  - *[added]* clues: no tachycardia with fever, normal CRP, fever only when unobserved
- **Drug fever:**
  - the commonest healthcare-associated FUO cause
  - stop or replace the suspected drug
  - *[added]* β-lactams, anticonvulsants, sulfonamides; fever typically resolves within 48–72 h of stopping

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6. Rapid Revision (Last-Day)

- ⭐ **Classic FUO:** > **38.3 °C (101 °F)**, > **3 weeks**, undiagnosed after **3 outpatient visits / 3 inpatient days**
- **Other FUO types:**
  - **nosocomial**: 3 days
  - **neutropenic**: ANC < 500, 3 days incl. 2 days culture
  - **HIV**: > 4 weeks outpatient / > 3 days inpatient

- **FUO causes:** infection 33 %, misc 29 %, malignancy 25 %, autoimmune 13 %
- **Healthcare-associated FUO: drug fever**, line infection, VTE, *C. diff*, abscess, transfusion, candidaemia
- **FUO approach:**
  - H&P → **stop antibiotics and steroids** → obligatory labs (**ESR/CRP, LDH, CK, ANA, RF, 3 blood cultures, CXR, abdominal US, TST/IGRA**)
  - → **exclude thermometer manipulation** → **exclude drug fever**
  - → PET-CT → biopsy (bone marrow / lymph node)
- **FUO treatment:**
  - **no empiric therapy** unless deteriorating
  - undiagnosed → specialist → trial of anti-TB / colchicine / steroids / **anakinra**
- **Malaria basics:** female ***Anopheles***; ~212 M cases, ~429 k deaths/yr
- **Malaria species:**
  - **vivax/ovale**: 48 h, **hypnozoites**
  - **malariae**: 72 h
  - **falciparum**: irregular, severe
  - **knowlesi**: 24 h
- **Life cycle:** sporozoite → liver schizont → merozoite → RBC trophozoite → schizont → rupture (fever) → gametocytes
- **Relapse / recrudescence / reinfection:**
  - **relapse** (vivax/ovale, hypnozoites, **primaquine after G6PD**)
  - **recrudescence** (persistent blood stage; falciparum/malariae; inadequate Rx or resistance)
  - **reinfection** (new bite)
- **Severe malaria (falciparum, cytoadherence):** cerebral, pulmonary oedema/ARDS, AKI, severe anaemia, **hypoglycaemia**, shock/DIC → **IV artesunate**, ICU
- **Diagnosis:**
  - **thick & thin smear** (gold standard), **×3, 12–24 h apart**; RDT faster but less sensitive
  - **Schüffner dots** = vivax/ovale; **crescent gametocytes** = falciparum
- **Malaria treatment:**
  - non-falciparum: **chloroquine** (+ **primaquine** for vivax/ovale)
  - falciparum: assume CQ resistance; quinine + doxy, quinine + SP, mefloquine, atovaquone/proguanil
  - monitoring: smear at **48 h, > 75 % drop**
  - **exchange transfusion > 10–15 %**
- **Prophylaxis:**
  - **DEET 30–35 %**, nets
  - CQ only in **Central America/Haiti/parts of Middle East**
  - **SE Asia: doxy or atovaquone/proguanil**
  - mefloquine ✗ seizures/psych
  - atovaquone/proguanil ✗ renal/pregnancy
  - **primaquine ✗ G6PD**
- **Typhoid:**
  - *S.* Typhi/Paratyphi A/B/C; **faecal–oral**; incubation **5–30 d**
  - **stepwise fever, relative bradycardia, rose spots, coated tongue**, constipation/diarrhoea, delirium
  - **blood culture (week 1)**; stool/urine later
  - **cipro or azithro**; **ceftriaxone** if quinolone-resistant

- week 3+: **Peyer's patch perforation/haemorrhage**, encephalopathy, myocarditis, cholecystitis, **osteomyelitis (sickle cell)**
- **carrier 1–5 %**, **gallbladder/gallstones**, shedding > 1 yr, "Typhoid Mary" → prolonged antibiotics ± cholecystectomy

- **Traveller's diarrhoea:**
  - **≥ 3 loose stools/day + 1 symptom**, within **1 week**, lasts **2–4 days**
  - **ETEC > 50 %**
  - **dysentery → Shigella/Entamoeba**
  - **fluids**; antibiotics (FQ/azithro/rifaximin) for severe; loperamide

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7. Exam-Style Questions

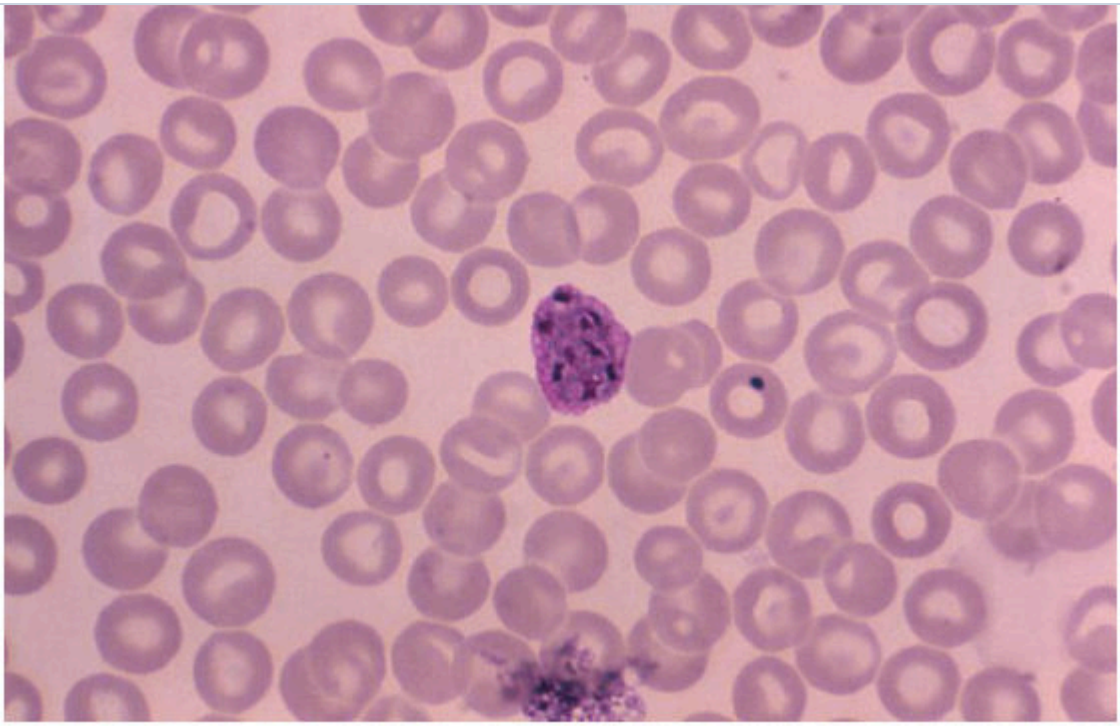
Q1. (Case Scenario 1) A 29-year-old woman returns from rural India after taking weekly chloroquine prophylaxis. She has fever 39.3 °C, jaundice, Hb 10 g/dL, and a smear with fragmented RBCs and ring-shaped intra-erythrocytic inclusions. Most likely cause? A. *P. ovale* · **B. *P. falciparum*** · C. Dengue virus · D. Chikungunya virus · E. Hepatitis A

> **Answer: B** *[reasoning added; key not on slides]*.
> - Ring forms inside RBCs = *Plasmodium*, which rules out C–E (viruses).
> - India has **chloroquine-resistant falciparum**, so prophylaxis failed.
> - Haemolysis with jaundice fits high parasitaemia.
> - *P. ovale* is chloroquine-sensitive and milder.

Q2. (Case Scenario 2) A 14-year-old girl returned from Indonesia 2 weeks ago. One week ago she had several days of low-grade fever and **constipation**. Now she has fever 39.3 °C, abdominal pain, profuse **non-bloody** diarrhoea, **hepatosplenomegaly** and a faint **salmon-coloured** maculopapular rash on the trunk. Causal organism? A. *E. coli* · B. *C. difficile* · C. *Shigella dysenteriae* · **D. *Salmonella* Typhi**

> **Answer: D** *[reasoning added; key not on slides]*.
> - Stepwise fever with early **constipation** then diarrhoea, plus **rose spots** (salmon-coloured truncal rash) and **hepatosplenomegaly**, in a traveller → typhoid.
> - The timeline (onset ~1 week after return) fits the 5–30-day incubation.
> - **Why not the others:**
> - ETEC (A) causes watery diarrhoea **within 1 week of arrival** without rash or organomegaly
> - *Shigella* (C) → **bloody** dysentery
> - *C. diff* (B) → antibiotic/healthcare exposure
> - **Next step: blood cultures**.

Q3. (Case Scenario 3) A 55-year-old woman has fever, chills, headache, nausea and scleral icterus. **Nine months ago** she had similar episodic fever after a trip to **Indonesia**, was treated with **chloroquine** and recovered. **No new travel.** The smear shows an enlarged parasitised RBC. Cause of recurrence?



Slide 39: Case 3 smear

A. Decline in antibodies · B. Reinfection · C. Natural drug resistance · **D. Reactivation of dormant liver stage** · E. Dissemination within macrophages

> **Answer: D** *[reasoning added; key not on slides]*.
>
> - This is **relapse** from ***P. vivax/ovale* hypnozoites**.
> - Chloroquine alone kills only the blood stage. **Primaquine (after G6PD)** was needed for radical cure.
> - B needs new exposure. C (resistance) would cause early **recrudescence**, not recurrence after 9 months symptom-free.
> - E describes *Leishmania/Histoplasma*, not *Plasmodium*.

Q4. A patient has 4 weeks of fever to 38.6 °C. Three outpatient visits found no cause. She was started on empiric ceftriaxone and prednisolone by another clinic, and she is haemodynamically stable. Best next step? A. Add vancomycin · **B. Stop antibiotics and glucocorticoids, then do the obligatory work-up** · C. PET-CT immediately · D. Bone-marrow biopsy · E. Trial of anakinra

> **Answer: B** (slides 10, 12).
>
> - Antibiotics and steroids **mask** the diagnosis.
> - The flowchart puts **stopping them before the obligatory labs**, including 3 blood cultures.
> - PET-CT and biopsy are **second tier**. Anakinra is a specialist trial for persistently undiagnosed FUO.
> - Empiric treatment is only for **rapid deterioration**.

Q5. A man treated for *P. vivax* is about to start primaquine. Which test must precede it? A. Hb electrophoresis · **B. G6PD level** · C. ECG for QT · D. Creatinine · E. HIV test

Answer: B (slides 17, 22, 23).

- Primaquine (and tafenoquine) causes **severe haemolysis** in G6PD deficiency.

Q6. A traveller with *P. falciparum* has a GCS of 9 and glucose of 2.4 mmol/L. Best treatment? A. Oral chloroquine · B. Oral atovaquone/proguanil · **C. IV artesunate + IV dextrose** · D. Primaquine · E. Doxycycline alone

Answer: C (slide 19, 22).

- Cerebral malaria and hypoglycaemia = **severe malaria**.
- **IV artesunate** is the drug of choice; correct glucose.

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8. Exam Pitfalls & Traps

Trap	Correct thinking
"FUO = any fever without a source"	❌ Needs **> 38.3 °C, > 3 weeks, and 3 visits / 3 inpatient days** of evaluation
38.3 °C = 100.1 °F (slide 6)	❌ **38.3 °C ≈ 101 °F** (slide 10 is correct)
Start empiric antibiotics/steroids in stable FUO	❌ Masks the diagnosis; **only if deteriorating or life-threatening**
PET-CT/biopsy as first step	❌ **Second tier** after the obligatory panel
Recurrent malaria months later = drug resistance	❌ No new travel + vivax/ovale = **relapse (hypnozoites)**
Relapse with *P. falciparum*	❌ Falciparum has **no hypnozoites**; recurrence = **recrudescence or reinfection**
Chloroquine + nothing else for vivax	❌ Add **primaquine** (after **G6PD**) for radical cure
Chloroquine prophylaxis is fine for India/Africa/SE Asia	❌ Only **Central America, Haiti, parts of the Middle East**
One negative smear excludes malaria	❌ **3 smears, 12–24 h apart**
RDT is the gold standard	❌ **Thick/thin smear** is; RDT is faster but **less sensitive**
Severe malaria = oral therapy	❌ **IV artesunate**; ICU; check **glucose**
Mefloquine for a traveller with epilepsy/depression	❌ Contraindicated → atovaquone/proguanil or doxy
Atovaquone/proguanil in pregnancy	❌ Avoid (slide); chloroquine is safe where effective
Typhoid: stool culture first	❌ **Blood culture in week 1**; stool/urine later
Fever + tachycardia always	❌ Typhoid → **relative bradycardia**
Typhoid carrier → just more antibiotics	⚠️ Prolonged antibiotics; **cholecystectomy if refractory** (gallstones)
Traveller's diarrhoea → stool culture + antibiotics routinely	❌ **Fluids**; tests only for systemic/prolonged illness
Bloody febrile diarrhoea → loperamide	❌ Dysentery (*Shigella/Entamoeba*); *[added]* avoid antimotility
Uncomplicated falciparum → quinine first line	⚠️ Slide lists quinine combos; *[added]* current guidance = **ACT (artemether–lumefantrine)**